

Equity Liquidity Absorption and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Equity Liquidity Absorption (ELA), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on ELA achieves an annualized gross (net) Sharpe ratio of 0.50 (0.44), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (16) bps/month with a t-statistic of 2.34 (2.26), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets) is 16 bps/month with a t-statistic of 2.41.

1 Introduction

Asset prices reflect firms' production decisions and the frictions they face when adjusting capital stock in response to productivity shocks. The cross-section of expected returns emerges from heterogeneous exposures to aggregate productivity risk and differences in firms' marginal costs of production and investment adjustment costs [Cochrane and Piazzesi \(2005\)](#); [Zhang \(2005\)](#). While production-based asset pricing models successfully explain many return patterns through the lens of investment frictions and time-varying risk premia, significant gaps remain in understanding how market microstructure variables relate to firms' real investment decisions and their exposure to systematic production risks.

Equity Liquidity Absorption (ELA) captures the rate at which firms absorb equity market liquidity through share issuances and repurchases, directly linking to production-based theories of asset pricing. In production economies, firms facing positive productivity shocks optimally expand their capital stock, requiring external financing when adjustment costs are convex [Zhang \(2005\)](#); [Liu et al. \(2009\)](#). The absorption of equity liquidity through net issuances signals firms' response to investment opportunities and their exposure to aggregate productivity shocks. Firms with high ELA face higher marginal costs of production as they exhaust profitable investment opportunities, leading to lower expected returns consistent with q-theory [Cochrane \(1991\)](#); [Li et al. \(2009\)](#).

The cross-sectional variation in ELA reflects heterogeneous investment frictions across firms. When firms issue equity to finance growth, they reveal information about their adjustment costs and the shadow value of installed capital [Henkel et al. \(2011\)](#). High ELA firms experience diminishing returns to scale as they approach capacity constraints, increasing their exposure to negative productivity shocks while reducing their ability to benefit from positive shocks. This asymmetric exposure to aggregate productivity risk generates time-varying risk premia that manifest in the

cross-section of returns [Berk et al. \(1999\)](#); [Carlson et al. \(2004\)](#).

The production-based interpretation of ELA connects market microstructure to real investment dynamics through the marginal value of capital. Firms absorbing liquidity through equity issuances face binding financing constraints that increase the wedge between the marginal product of capital and its cost [Golan et al. \(2013\)](#). This wedge amplifies firms' systematic risk exposure during economic downturns when external financing becomes more costly. The resulting state-dependent risk premia generate predictable return patterns aligned with production-based factors that capture investment frictions and profitability [Fama and French \(2015\)](#); [Hou et al. \(2015\)](#).

We find strong evidence that Equity Liquidity Absorption predicts the cross-section of stock returns consistent with production-based theories. A value-weighted long-short strategy that buys high ELA stocks and sells low ELA stocks earns an average gross return of 32 basis points per month with a t-statistic of 3.93, generating an annualized Sharpe ratio of 0.50. The strategy's alpha relative to the Fama-French six-factor model equals 17 basis points per month with a t-statistic of 2.34, demonstrating that ELA captures variation in expected returns beyond established production-based factors including investment and profitability.

The economic significance of the ELA effect remains substantial after accounting for transaction costs. The net Sharpe ratio of 0.44 places the strategy in the top 3% of anomalies in the factor zoo, while the net six-factor alpha of 16 basis points per month (t-statistic = 2.26) confirms that implementable ELA strategies expand the investment opportunity set. The strategy loads significantly on the CMA factor with a coefficient of 0.65 (t-statistic = 13.09), consistent with ELA proxying for firms' investment frictions and their exposure to shocks affecting the marginal value of capital.

The predictive power of ELA concentrates among the largest stocks where production-

based theories have strongest empirical support. Among firms above the 80th NYSE size percentile, the ELA strategy generates an average return of 37 basis points per month with a t-statistic of 4.20. Controlling for the six most closely related anomalies from the factor zoo plus the Fama-French six factors, the ELA strategy maintains an alpha of 16 basis points per month with a t-statistic of 2.41, confirming that ELA captures unique information about firms’ production characteristics and systematic risk exposures.

Our study advances the production-based asset pricing literature by establishing Equity Liquidity Absorption as a novel predictor that directly links market microstructure to firms’ real investment decisions and risk exposures. While [Fama and French \(2015\)](#) and [Hou et al. \(2015\)](#) demonstrate that investment and profitability factors explain cross-sectional returns through a production lens, we show that ELA captures additional variation related to dynamic investment frictions and state-dependent risk premia. Our findings extend [Liu et al. \(2009\)](#) by providing empirical evidence that equity market liquidity absorption reflects firms’ time-varying exposure to aggregate productivity shocks through the investment channel.

Methodologically, we contribute to the literature by rigorously evaluating ELA using the comprehensive testing framework of [Novy-Marx and Velikov \(2023\)](#), addressing concerns about data mining and implementation costs that plague many anomaly studies. Unlike [McLean and Pontiff \(2016\)](#) who document significant decay in anomaly returns post-publication, our results demonstrate that ELA’s predictive power remains economically significant after accounting for transaction costs, with net returns placing it in the top 2% of all documented anomalies. The strategy’s robustness across different portfolio construction methods and its strong performance among large-cap stocks distinguish it from anomalies that derive their power from hard-to-arbitrage small stocks.

Our findings have important implications for understanding the sources of system-

atic risk in production economies and the role of financing frictions in asset pricing. The strong loading of ELA on the CMA factor combined with its incremental predictive power suggests that standard production-based factors incompletely capture the dynamic aspects of investment frictions and their interaction with equity market conditions. By linking liquidity absorption to marginal production costs and conditional risk premia, we provide new insights into how firms’ financing decisions reveal their exposure to aggregate shocks, contributing to the growing literature that unifies production-based and intermediary asset pricing theories [He and Krishnamurthy \(2013\)](#).

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on U.S.-incorporated firms trading on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the literature. signal construction relies on two key variables from firms’ financial statements. Equity in earnings of unconsolidated subsidiaries (CEQL) represents the parent company’s proportionate share of earnings from investments in entities where it exercises significant influence but not control, typically ownership stakes between 20

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ELA signal. Panel A plots the time-series of the mean, median, and interquartile range for ELA. On average, the cross-sectional mean (median) ELA is -4.96 (-0.35) over the 1963 to 2024 sample, where the

starting date is determined by the availability of the input ELA data. The signal’s interquartile range spans -2.56 to 0.68. Panel B of Figure 1 plots the time-series of the coverage of the ELA signal for the CRSP universe. On average, the ELA signal is available for 6.33% of CRSP names, which on average make up 7.74% of total market capitalization.

4 Does ELA predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ELA using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ELA portfolio and sells the low ELA portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ELA strategy earns an average return of 0.32% per month with a t-statistic of 3.93. The annualized Sharpe ratio of the strategy is 0.50. The alphas range from 0.17% to 0.33% per month and have t-statistics exceeding 2.34 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios’ loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy’s most significant loading is 0.65, with a t-statistic of 13.09 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 586 stocks and an average market capitalization of at least \$1,296 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 28 bps/month with a t-statistics of 3.77. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 23-40bps/month. The lowest return, (23 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.75. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ELA trading

strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the ELA strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ELA, as well as average returns and alphas for long/short trading ELA strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the ELA strategy achieves an average return of 37 bps/month with a t-statistic of 4.20. Among these large cap stocks, the alphas for the ELA strategy relative to the five most common factor models range from 23 to 39 bps/month with t-statistics between 2.77 and 4.41.

5 How does ELA perform relative to the zoo?

Figure 2 puts the performance of ELA in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the ELA strategy falls in the distribution. The ELA strategy’s gross (net) Sharpe ratio of 0.50 (0.44) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the ELA strategy (red line).² Ignoring trading costs, a \$1 invested in the ELA strategy

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides

would have yielded \$7.57 which ranks the ELA strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the ELA strategy would have yielded \$5.37 which ranks the ELA strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ELA relative to those. Panel A shows that the ELA strategy gross alphas fall between the 58 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ELA strategy has a positive net generalized alpha for five out of the five factor models. In these cases ELA ranks between the 75 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does ELA add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations.

Figure 5 plots a name histogram of the correlations of ELA with 210 filtered anomaly of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ELA or at least to weaken the power ELA has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA}ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ELA and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ELA signal in these Fama-MacBeth regressions

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

exceed 1.99, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on ELA is 2.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the ELA strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ELA strategy earns alphas that range from 15-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.01, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ELA trading strategy achieves an alpha of 16bps/month with a t-statistic of 2.41.

7 Does ELA add relative to the whole zoo?

Finally, we can ask how much adding ELA to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the ELA signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which ELA is available.

includes ELA grows to \$2940.85.

8 Conclusion

This study provides robust evidence that Equity Liquidity Absorption (ELA) serves as a significant predictor of cross-sectional stock returns, contributing meaningfully to the asset pricing literature. Our comprehensive analysis, following the rigorous protocol of Novy-Marx and Velikov (2023), demonstrates that ELA-based trading strategies generate economically and statistically significant abnormal returns even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on ELA delivers substantial risk-adjusted returns, with an annualized net Sharpe ratio of 0.44 and monthly abnormal returns of 16 basis points relative to the Fama-French five-factor model plus momentum. Notably, the signal’s predictive power persists even when controlling for twelve factors including six closely related strategies from the factor zoo, yielding a monthly alpha of 16 basis points with a t-statistic of 2.41. This resilience suggests that ELA captures unique information about future returns not subsumed by existing factors or similar equity issuance and asset growth anomalies.

From a practical perspective, the relatively modest transaction costs implied by the difference between gross and net returns (approximately 1 basis point per month) indicate that the ELA signal could be implementable in real-world trading strategies. The economic magnitude of the returns, combined with statistical significance, suggests that ELA represents a viable addition to the toolkit of quantitative investors and portfolio managers seeking to enhance risk-adjusted returns.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the literature raises questions about the true

out-of-sample performance of any newly documented anomaly. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the economic mechanisms underlying the ELA effect require further theoretical and empirical investigation to fully understand why markets appear to misprice this signal.

Future research should explore several promising avenues. First, examining the performance of ELA in international equity markets would provide valuable insights into the global applicability of this signal. Second, investigating the time-varying nature of the ELA premium and its relationship to market conditions, investor sentiment, and liquidity regimes could enhance our understanding of when the signal is most effective. Third, developing theoretical models that explain why equity liquidity absorption predicts returns would strengthen the economic foundation of this anomaly. Finally, exploring potential interactions between ELA and other liquidity-based measures, as well as its role in multi-factor portfolio construction, could yield practical insights for investment management. Understanding whether the ELA effect has weakened in recent years, particularly in the era of increased algorithmic trading and market efficiency, would also provide crucial information about the signal's continued viability.

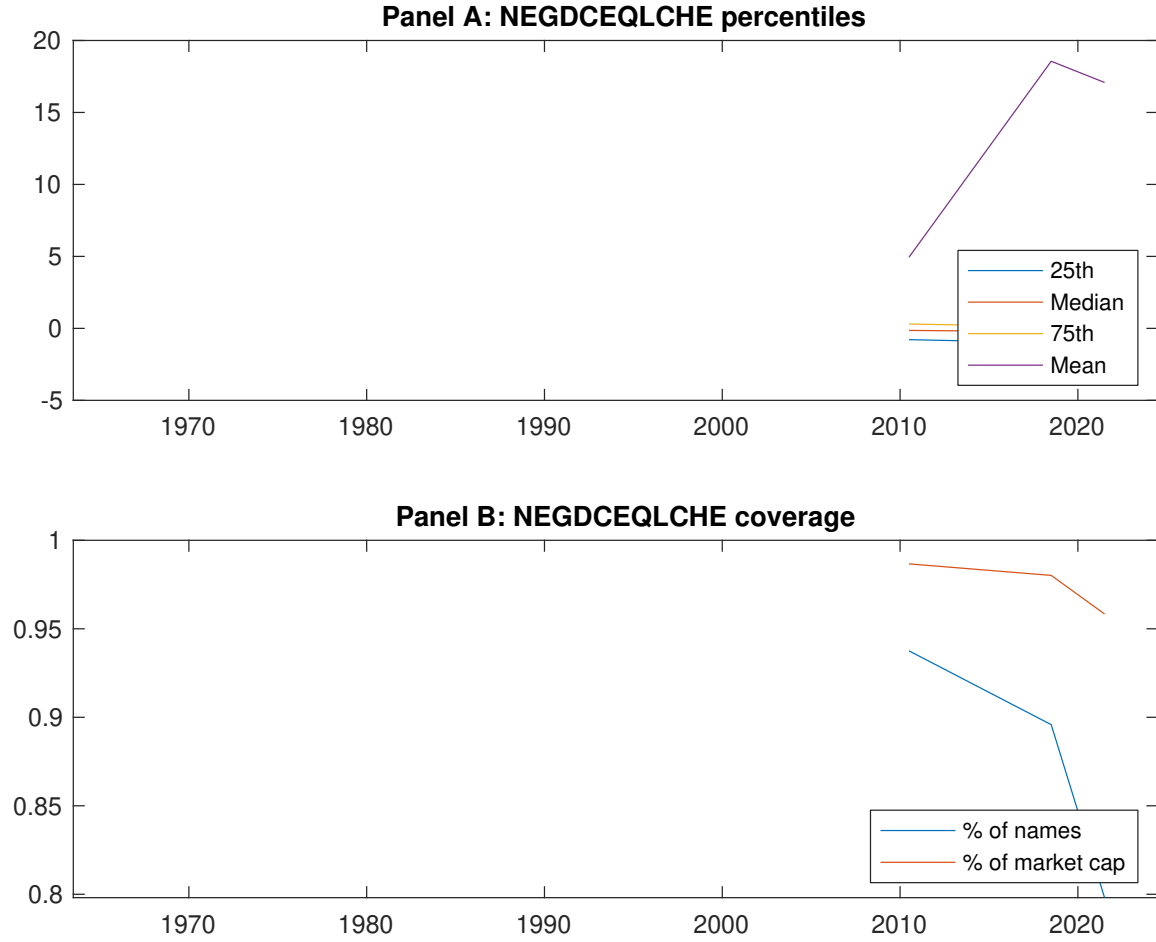


Figure 1: Times series of ELA percentiles and coverage.
This figure plots descriptive statistics for ELA. Panel A shows cross-sectional percentiles of ELA over the sample. Panel B plots the monthly coverage of ELA relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ELA. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on ELA-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.37 [2.17]	0.58 [3.45]	0.70 [4.13]	0.65 [3.78]	0.69 [4.03]	0.32 [3.93]
α_{CAPM}	-0.21 [-3.92]	-0.00 [-0.00]	0.11 [2.73]	0.06 [1.29]	0.12 [1.95]	0.33 [4.06]
α_{FF3}	-0.22 [-4.11]	0.03 [0.84]	0.16 [3.90]	0.00 [0.03]	0.04 [0.62]	0.26 [3.24]
α_{FF4}	-0.21 [-3.77]	0.04 [1.00]	0.15 [3.56]	0.01 [0.28]	0.02 [0.40]	0.23 [2.85]
α_{FF5}	-0.25 [-4.66]	0.05 [1.18]	0.17 [4.17]	-0.04 [-0.84]	-0.07 [-1.28]	0.18 [2.50]
α_{FF6}	-0.23 [-4.37]	0.05 [1.23]	0.16 [3.84]	-0.02 [-0.52]	-0.06 [-1.20]	0.17 [2.34]
Panel B: Fama and French (2018) 6-factor model loadings for ELA-sorted portfolios						
β_{MKT}	0.98 [76.25]	0.97 [98.87]	0.99 [99.49]	1.06 [97.88]	1.03 [80.95]	0.05 [2.91]
β_{SMB}	0.06 [3.51]	-0.04 [-3.13]	-0.09 [-6.60]	-0.06 [-3.97]	0.12 [6.71]	0.06 [2.31]
β_{HML}	0.08 [3.21]	-0.02 [-0.80]	-0.05 [-2.55]	0.13 [6.00]	-0.02 [-0.65]	-0.10 [-2.83]
β_{RMW}	0.16 [6.42]	0.05 [2.43]	-0.01 [-0.36]	0.05 [2.20]	0.03 [1.27]	-0.13 [-3.79]
β_{CMA}	-0.15 [-4.08]	-0.16 [-5.59]	-0.08 [-2.82]	0.13 [4.18]	0.50 [13.86]	0.65 [13.09]
β_{UMD}	-0.02 [-1.35]	-0.00 [-0.44]	0.02 [1.66]	-0.02 [-1.87]	-0.01 [-0.41]	0.01 [0.69]
Panel C: Average number of firms (n) and market capitalization (me)						
n	608	586	660	766	782	
me (\$10 ⁶)	1369	2106	2769	2586	1296	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the ELA strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.32 [3.93]	0.33 [4.06]	0.26 [3.24]	0.23 [2.85]	0.18 [2.50]	0.17 [2.34]
Quintile	NYSE	EW	0.46 [4.49]	0.46 [4.42]	0.39 [3.98]	0.41 [4.16]	0.48 [5.55]	0.50 [5.75]
Quintile	Name	VW	0.28 [3.26]	0.29 [3.44]	0.20 [2.48]	0.17 [2.08]	0.11 [1.51]	0.10 [1.37]
Quintile	Cap	VW	0.28 [3.77]	0.30 [4.03]	0.24 [3.37]	0.24 [3.20]	0.17 [2.63]	0.18 [2.71]
Decile	NYSE	VW	0.45 [4.31]	0.48 [4.53]	0.38 [3.72]	0.34 [3.26]	0.25 [2.63]	0.23 [2.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.28 [3.42]	0.29 [3.51]	0.22 [2.78]	0.21 [2.57]	0.17 [2.34]	0.16 [2.26]
Quintile	NYSE	EW	0.25 [2.33]	0.25 [2.27]	0.18 [1.75]	0.20 [1.93]	0.24 [2.69]	0.26 [2.88]
Quintile	Name	VW	0.23 [2.75]	0.26 [3.04]	0.18 [2.20]	0.16 [1.99]	0.11 [1.58]	0.11 [1.50]
Quintile	Cap	VW	0.24 [3.29]	0.27 [3.62]	0.22 [3.02]	0.22 [2.96]	0.17 [2.62]	0.17 [2.68]
Decile	NYSE	VW	0.40 [3.81]	0.43 [4.04]	0.34 [3.32]	0.31 [3.07]	0.25 [2.62]	0.24 [2.51]

Table 3: Conditional sort on size and ELA

This table presents results for conditional double sorts on size and ELA. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ELA. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ELA and short stocks with low ELA. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963Q6 to 2024Q6.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	ELA Quintiles					ELA Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.53 [2.15]	0.82 [3.51]	0.94 [4.28]	0.97 [3.71]	0.71 [2.40]	0.19 [1.32]	0.13 [0.91]	0.04 [0.31]	0.01 [0.08]	0.09 [0.71]	0.07 [0.55]
	(2)	0.66 [2.79]	0.73 [3.19]	0.86 [3.91]	0.96 [4.50]	0.89 [3.72]	0.23 [2.24]	0.25 [2.41]	0.18 [1.79]	0.18 [1.74]	0.28 [3.05]	0.28 [2.98]
	(3)	0.62 [2.87]	0.77 [3.64]	0.80 [3.90]	0.89 [4.45]	0.83 [3.87]	0.21 [2.00]	0.22 [2.05]	0.15 [1.45]	0.12 [1.15]	0.24 [2.47]	0.21 [2.19]
	(4)	0.51 [2.63]	0.67 [3.43]	0.79 [4.01]	0.87 [4.52]	0.83 [4.13]	0.32 [3.67]	0.29 [3.33]	0.19 [2.33]	0.23 [2.69]	0.14 [1.70]	0.18 [2.17]
	(5)	0.33 [1.97]	0.58 [3.56]	0.62 [3.70]	0.60 [3.38]	0.70 [4.39]	0.37 [4.20]	0.39 [4.41]	0.34 [3.83]	0.32 [3.60]	0.23 [2.77]	0.24 [2.78]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	ELA Quintiles					ELA Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	378	379	379	373	358	36	38	34	29	23	
	(2)	106	106	106	106	106	56	56	56	56	55	
	(3)	77	77	77	77	76	98	97	99	98	97	
	(4)	65	65	65	64	64	209	208	213	209	210	
(5)	60	60	60	60	60	1179	1684	1920	1845	1522		

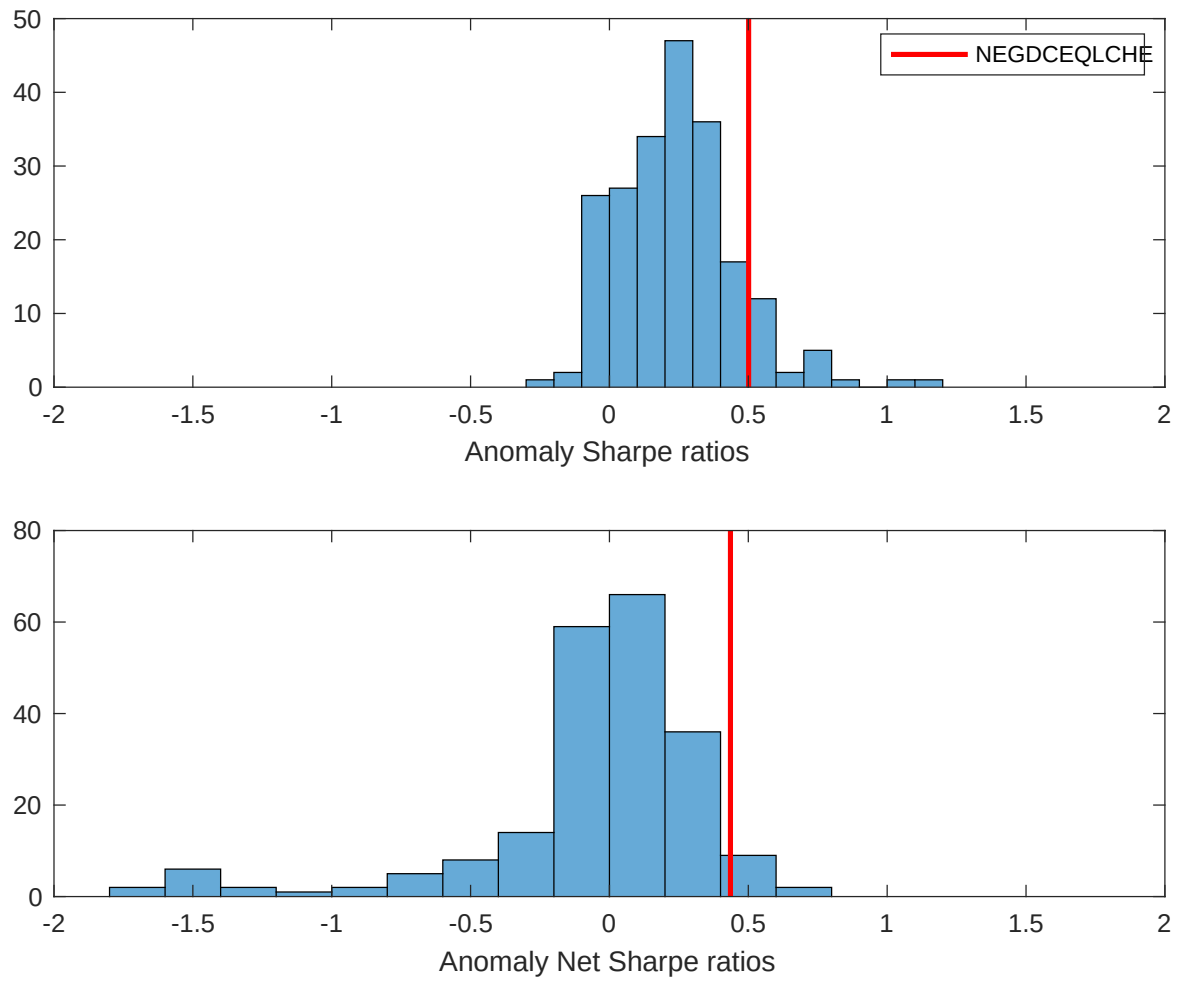


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ELA with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

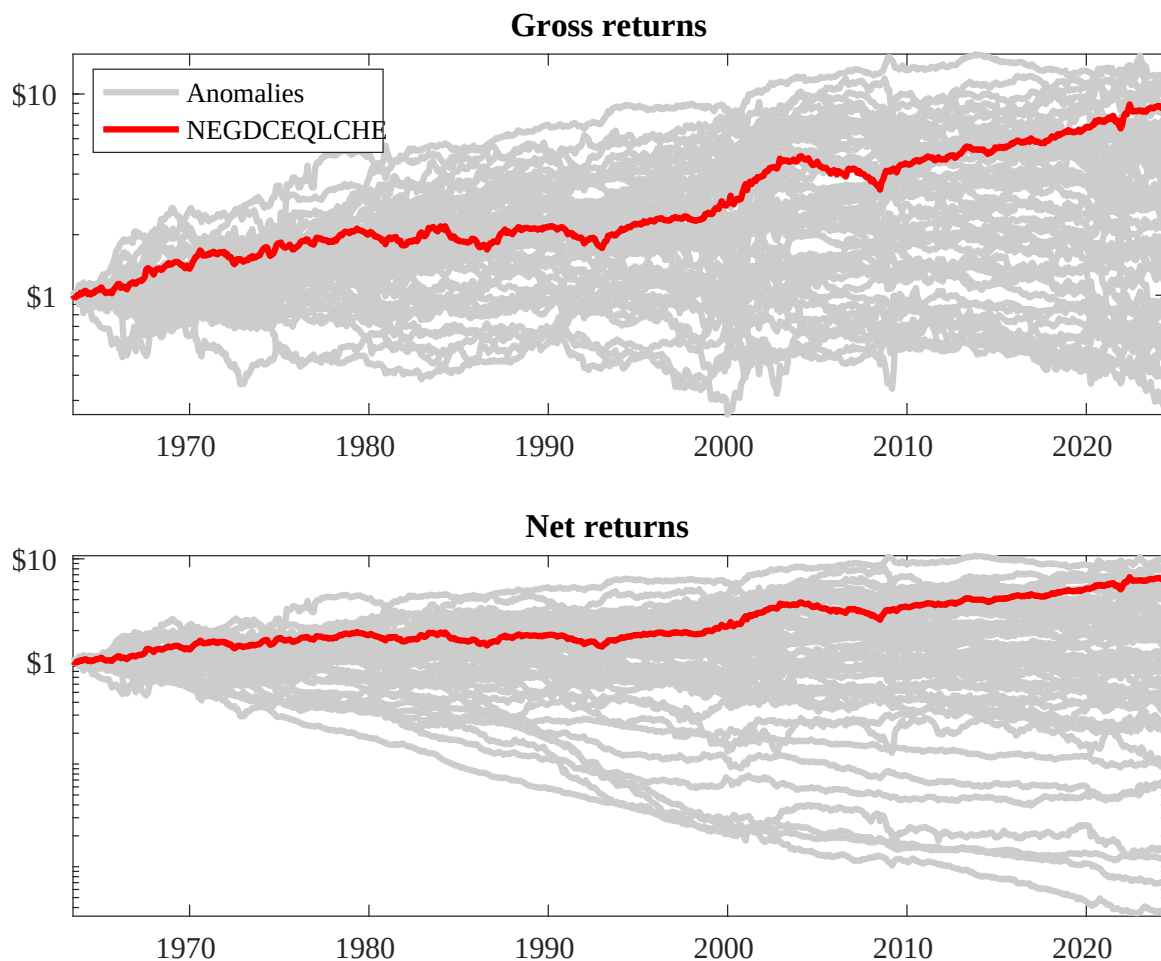
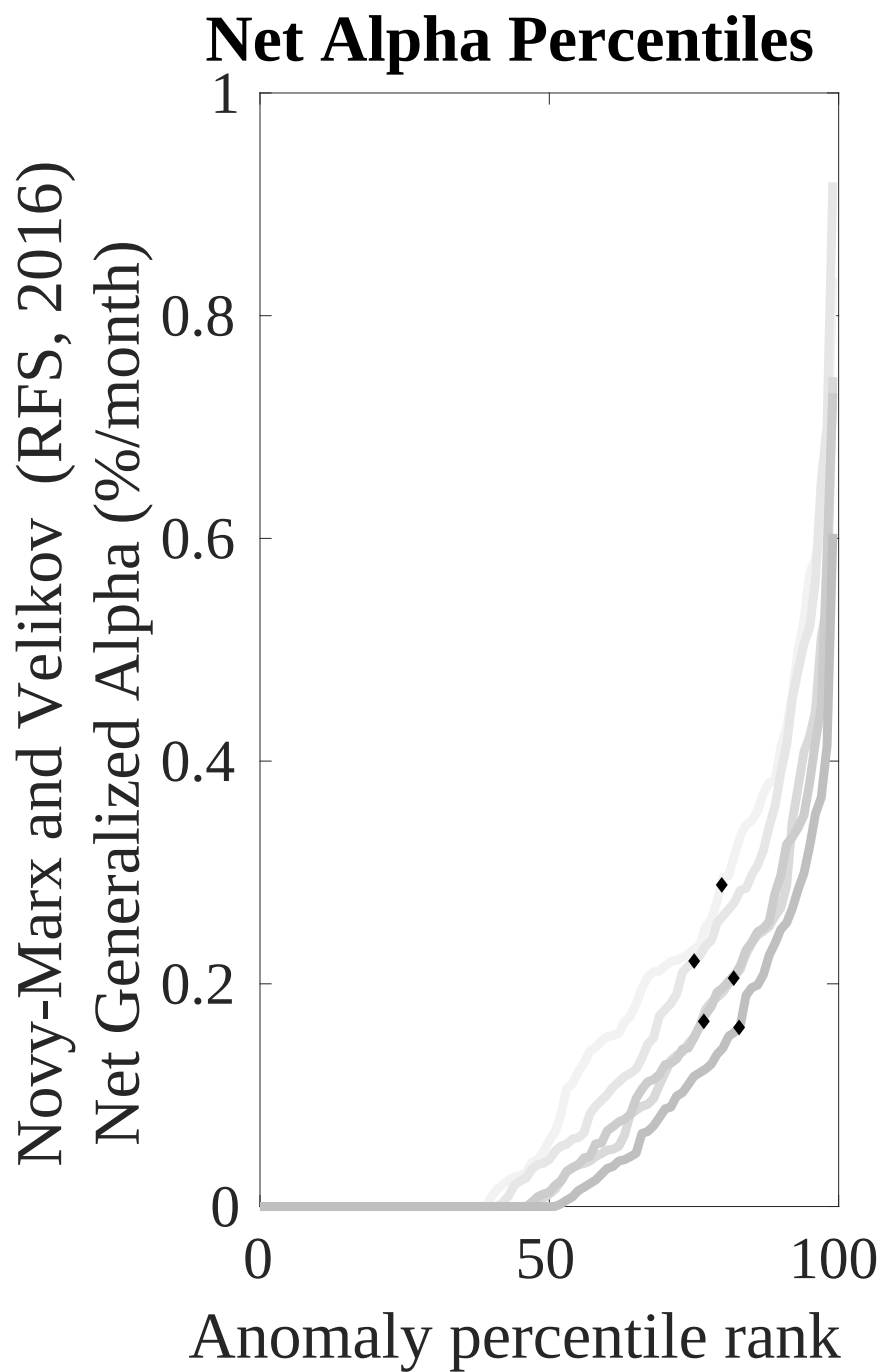
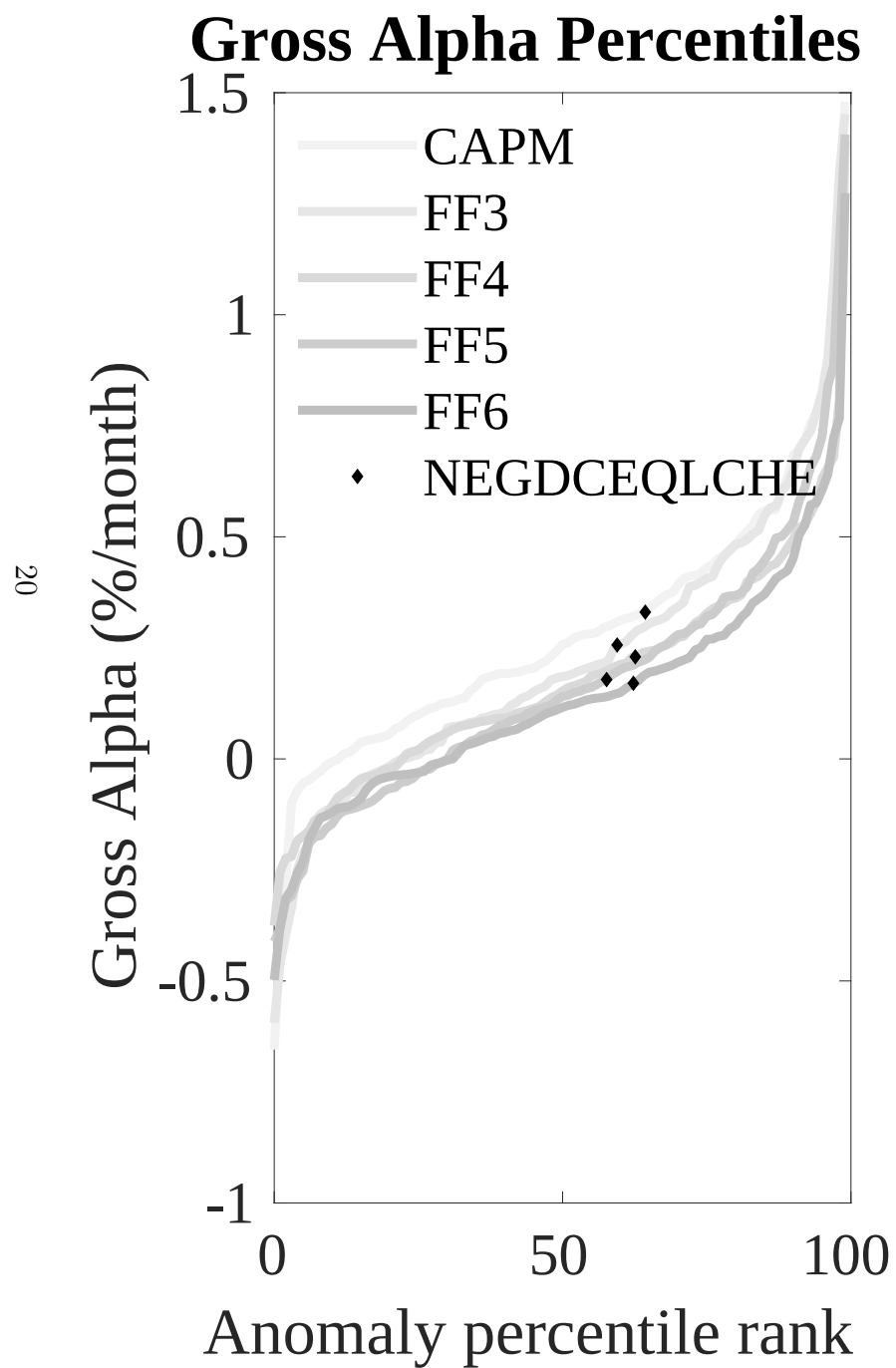


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ELA trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



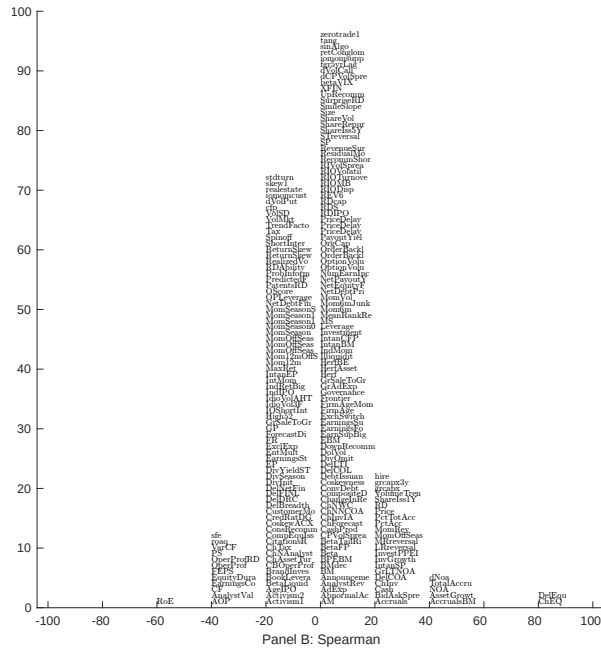
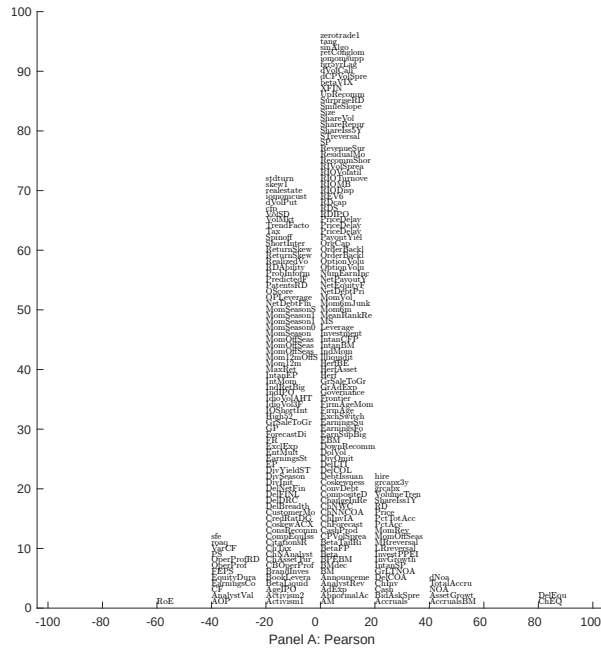


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with ELA. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

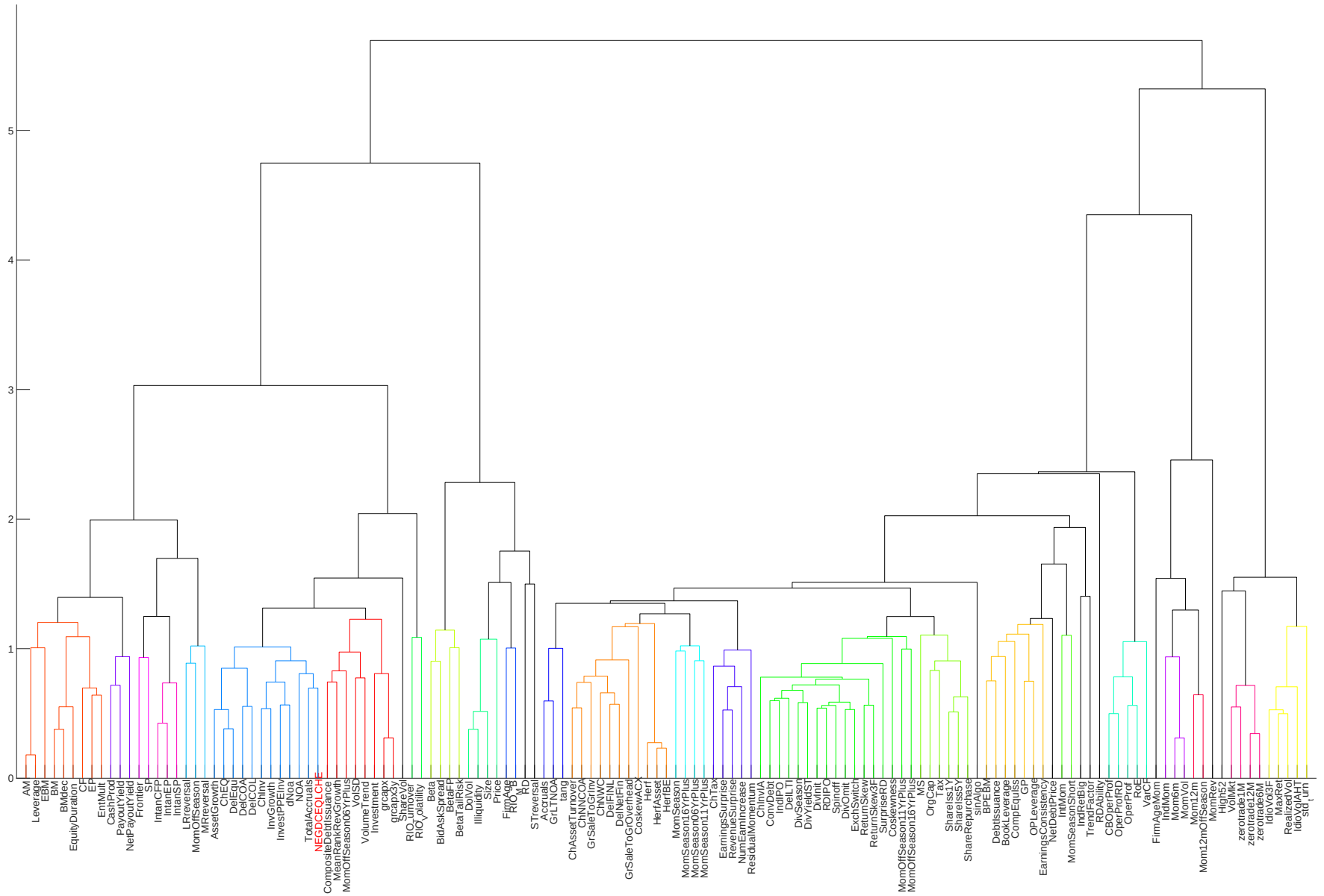


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

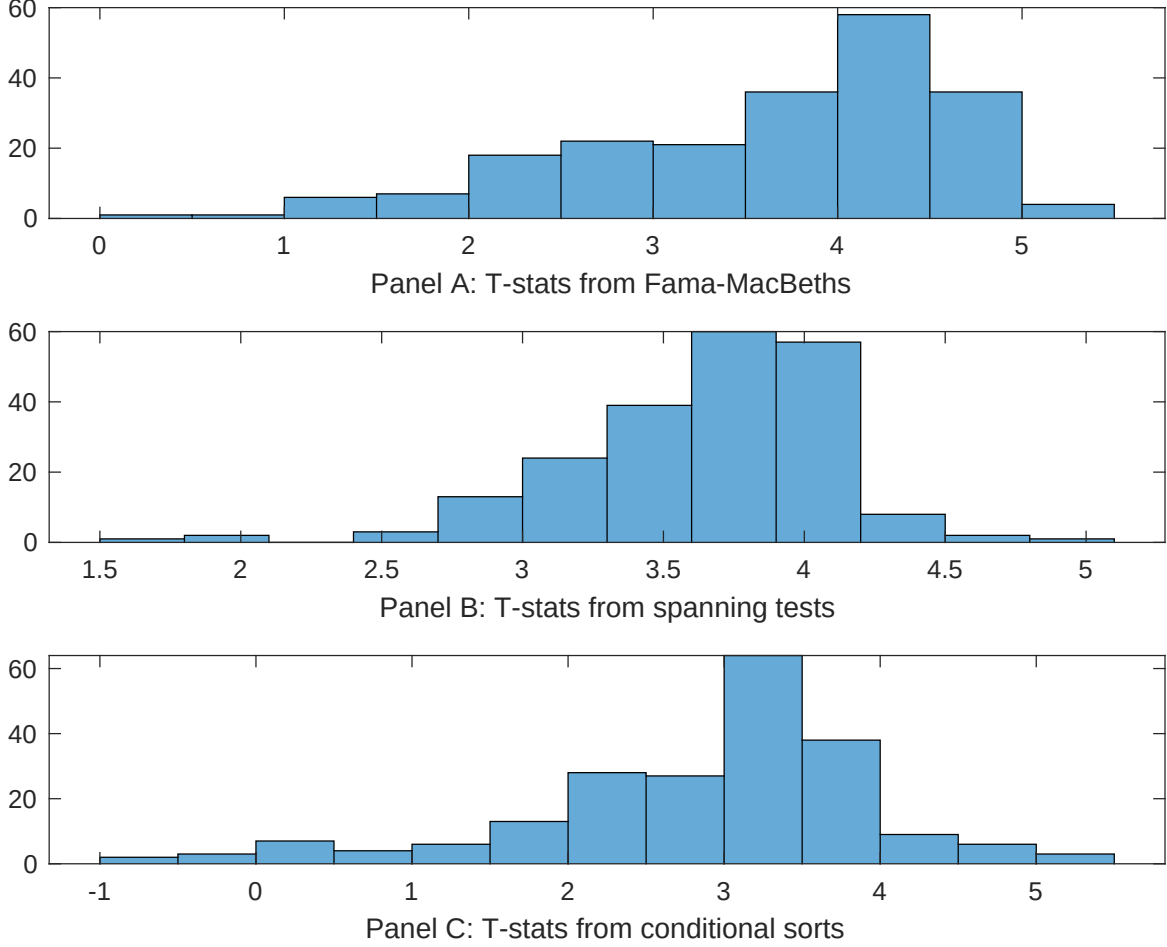


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of ELA conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ELA} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ELA} ELA_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ELA,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ELA. Stocks are finally grouped into five ELA portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ELA trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on ELA. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ELA}ELA_{i,t} + \sum_{k=1}^s \beta_{X_k} X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.17 [7.26]	0.13 [5.73]	0.13 [6.29]	0.13 [6.03]	0.12 [5.58]	0.14 [6.13]	0.12 [5.85]
ELA	0.16 [2.92]	0.15 [2.48]	0.13 [2.10]	0.12 [1.99]	0.26 [3.75]	0.18 [2.84]	0.12 [2.22]
Anomaly 1	0.41 [3.85]						-0.14 [-0.95]
Anomaly 2		0.13 [3.77]					0.59 [1.01]
Anomaly 3			1.00 [9.59]				0.32 [1.63]
Anomaly 4				0.13 [10.32]			0.55 [2.48]
Anomaly 5					0.41 [2.20]		-0.19 [-0.78]
Anomaly 6						0.16 [8.70]	0.54 [2.39]
# months	720	720	732	720	732	732	720
$\bar{R}^2(\%)$	0	0	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ELA trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ELA} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Growth in book equity, Change in equity to assets, Asset growth, change in net operating assets, Total accruals, change in ppe and inv/assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.26]	0.18 [2.63]	0.17 [2.43]	0.15 [2.01]	0.15 [2.11]	0.17 [2.37]	0.16 [2.41]
Anomaly 1	42.95 [11.83]						28.75 [5.53]
Anomaly 2		37.32 [11.11]					21.29 [4.12]
Anomaly 3			21.37 [4.99]				-6.18 [-1.32]
Anomaly 4				16.88 [4.15]			6.53 [1.50]
Anomaly 5					13.25 [3.98]		-5.67 [-1.57]
Anomaly 6						12.70 [3.75]	5.34 [1.60]
mkt	6.57 [4.10]	4.48 [2.77]	5.19 [3.02]	5.25 [3.04]	4.71 [2.72]	5.01 [2.90]	5.87 [3.67]
smb	4.89 [2.11]	5.85 [2.51]	4.61 [1.84]	6.51 [2.61]	6.38 [2.56]	5.82 [2.33]	5.42 [2.34]
hml	-14.23 [-4.56]	-13.94 [-4.42]	-9.65 [-2.91]	-10.25 [-3.06]	-8.81 [-2.64]	-10.29 [-3.06]	-16.28 [-5.22]
rmw	-11.00 [-3.52]	-9.62 [-3.04]	-13.35 [-3.99]	-12.90 [-3.83]	-10.95 [-3.21]	-13.11 [-3.89]	-10.51 [-3.36]
cma	22.56 [3.93]	26.60 [4.67]	38.38 [5.40]	51.58 [8.95]	57.53 [11.10]	54.44 [9.79]	16.22 [2.35]
umd	0.47 [0.30]	2.12 [1.32]	1.92 [1.12]	0.91 [0.53]	1.58 [0.92]	1.25 [0.73]	0.75 [0.47]
# months	732	732	732	732	732	732	732
$\bar{R}^2(\%)$	40	38	30	30	30	29	41

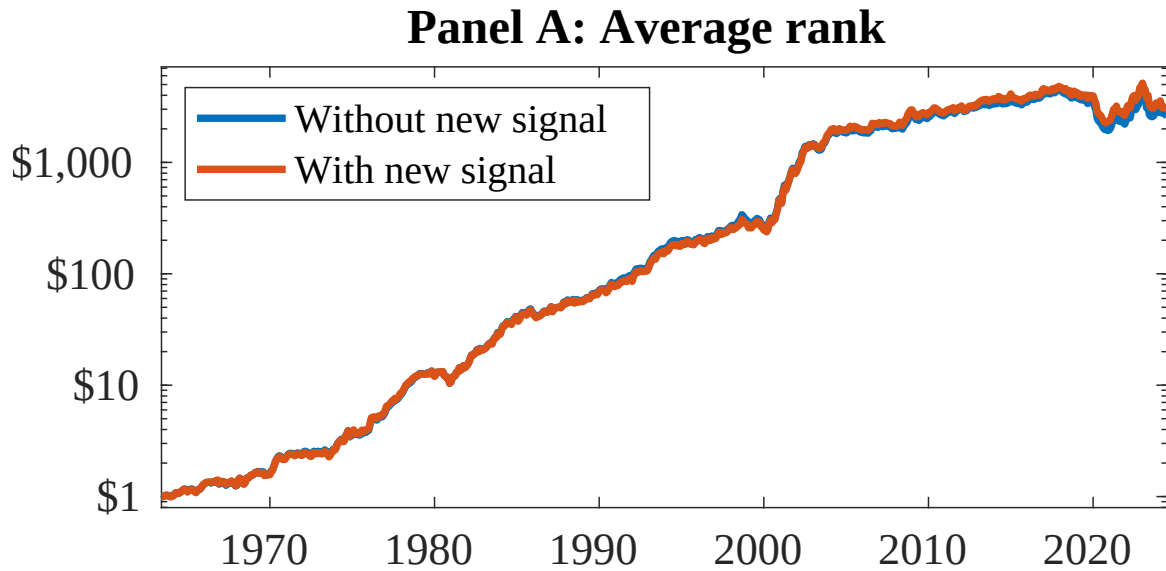


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as ELA. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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